Lesson 12A Equilibrium

Equilibrium

In physics, we often analyze situations where objects remain steady, even though forces may be acting on them. For example:

- A **book** resting on a table,
- A **puck** gliding with constant velocity on ice,
- A **ceiling fan** rotating at a steady rate,
- A **bicycle wheel** rolling smoothly at constant velocity.

Static Equilibrium

In this chapter, we focus primarily on systems that are not moving at all.

These systems are said to be in **static equilibrium**. Of the earlier examples, only the **book at rest** on the table is in static equilibrium.

- If a system returns to equilibrium after a small disturbance, it is in **stable static equilibrium**.
- If a slight displacement leads to departure from equilibrium, it is in **unstable static equilibrium**.

Conditions for Equilibrium

Equilibrium conditions are derived from Newton's second law applied to both linear and rotational motion.

Force Equilibrium:

$$\sum \vec{F} = 0$$

This means the **net force** on the body is zero, and the object's center of mass does not accelerate.

Torque Equilibrium:

$$\sum \vec{\tau} = 0$$

This means the **net torque** about any point is zero, and the object does not experience angular acceleration.

For equilibrium in the **xy-plane**, these conditions simplify to:

$$\sum F_x = 0 \ , \ \sum F_y = 0 \ , \ \sum \tau_z = 0$$

The Gravitational Force and the Center of Gravity

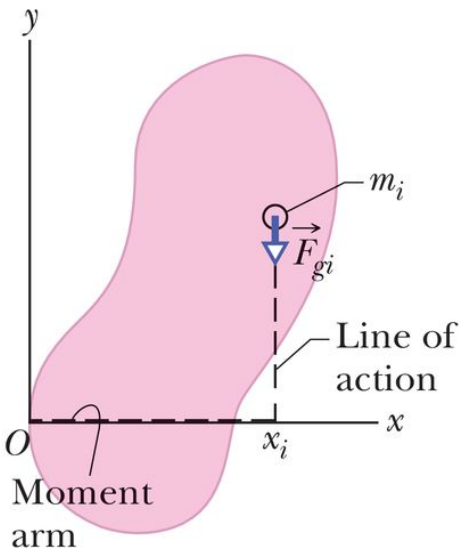
The total gravitational force on a body is the sum of the gravitational forces acting on all its individual parts:

$$\vec{F}_g = \sum \vec{F}_{g_i}$$

If the gravitational field is uniform (which it usually is near Earth's surface), we can treat the entire force as acting at a single point: the **center of gravity** (often approximated by the center of mass).

Center of Gravity and Torque

When calculating the gravitational torque on an extended object, we can consider the gravitational forces acting on each mass element of the object.



Each mass element m_i experiences a gravitational force $\vec{F}_{gi} = m_i \vec{g}$, which acts vertically downward. The torque due to each of these forces about a chosen origin O is:

$$\tau_i = x_i F_{gi}$$

where x_i is the perpendicular distance (moment arm) from the axis at O to the line of action of the force $\vec{F}_{gi}$. The total gravitational torque is the sum over all elements:

$$\sum \tau_i = \sum x_i F_{gi} = \sum x_i m_i g$$

Since g is constant, it factors out:

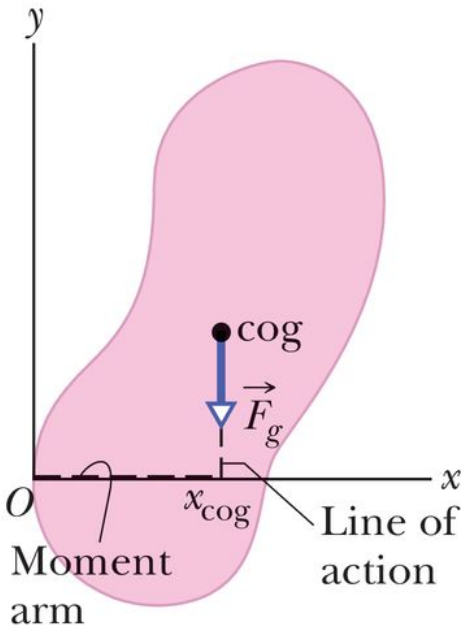
$$\sum \tau_i = g \sum x_i m_i$$

We can express this in terms of the total gravitational force $F_g = Mg$, where $M = \sum m_i$ is the total mass. Then we define the **center of gravity** x_{cog} such that:

$$x_{\text{cog}} = \frac{1}{M} \sum x_i m_i$$

This allows us to write the total torque due to gravity as if the entire weight Mg acts at a single point:

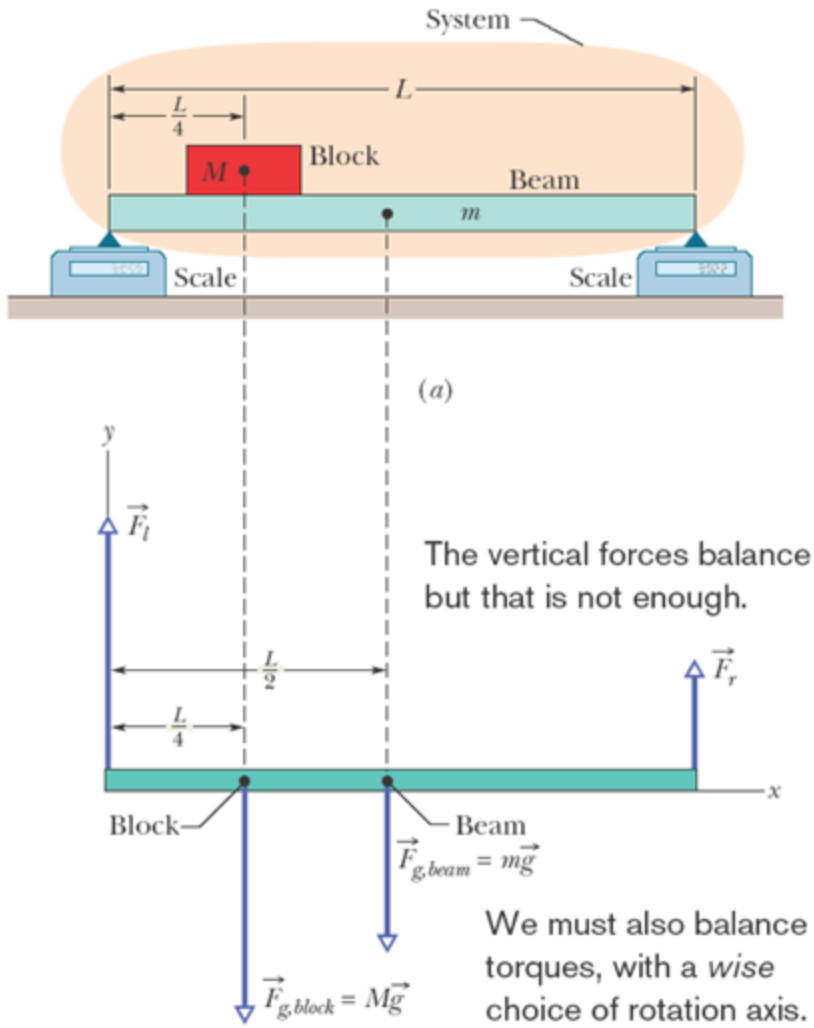
$$\tau = x_{\text{cog}} M g$$



Thus, the torque due to gravity is the same as if the force $\vec{F}_g$ acts at the center of gravity of the object. This is shown in the figures above: the distributed torques from all mass elements (first image) can be replaced with a single equivalent torque from the total weight acting at x_{cog} (second image).

Example: Balancing a Horizontal Beam

A uniform beam of length L and mass m=1.8 kg is at rest on two scales. A uniform block, with mass M=2.7 kg is at rest on the team, with its center a distance L/4 from the beam's left end. What do the scales read?



Given: $M = 2.7\,\mathrm{kg},\ m = 1.8\,\mathrm{kg}$

- Set the rotation axis at $x = 0$
- Sum of torques gives:

$$\frac{1}{4}MgL + \frac{1}{2}mgL = F_r L \Rightarrow F_r = 15\,\mathrm{N}$$

- Vertical force balance:

$$F_l = (M+m)g - F_r = 29\,\mathrm{N}$$

Hewitt-Drew-it! PHYSICS 41.Torques on a Plank